QUAKELOGIC

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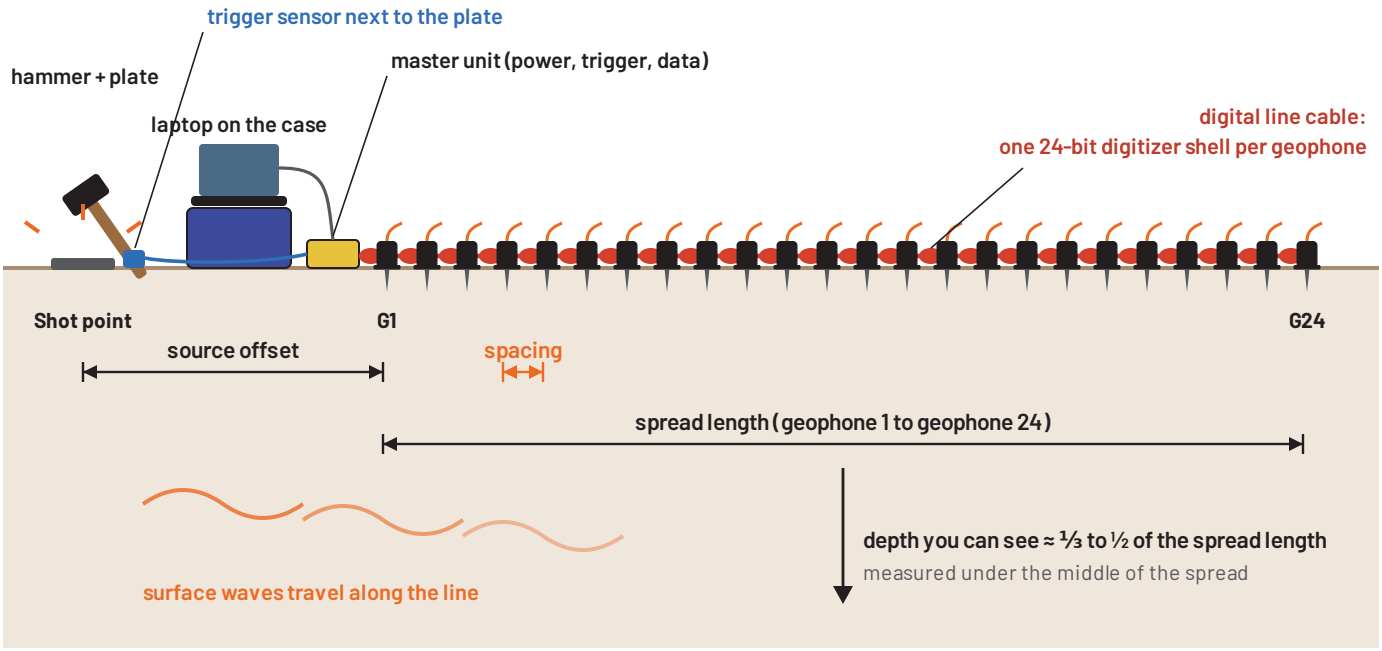
MASW FIELD GUIDE

MASW STUDIO

SET UP · RECORD · ANALYSE

Equipment, line layout, seismograph settings and MASW Studio, step by step, for your first survey.

MASW STUDIO 1.3.2 · WINDOWS 11



FIELD GUIDE

From the equipment case to a Vs report

FOR FIRST-TIME MASW USERS · MASW STUDIO 1.3.2

1 Before you start
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1. Before you start

This guide takes you from the equipment case to a finished shear-wave velocity (Vs) report. It follows the setup shown in QuakeLogic's seismograph video, step by step, and assumes no previous experience. Read it once in the office, then take the checklists (section 10) to the field.

What MASW measures

When a hammer hits the ground, it sends out surface waves that roll along the ground like ripples on water. Their low frequencies (long waves) reach deep into the ground, and their high frequencies (short waves) stay shallow. Each frequency therefore travels at the speed of the ground it passes through.

A line of geophones records these waves. MASW Studio measures the speed of each frequency and turns those speeds into a profile of shear-wave velocity (Vs) against depth. Vs tells an engineer how stiff the ground is. It is used for seismic site class (Vs30), foundation design, compaction checks and finding soft zones or voids.

> **You need three things to be right in the field:** a straight line of well-planted geophones at a measured spacing, a clean trigger at the moment of each hit, and the positions entered in the recording software. Everything else can be fixed in MASW Studio afterwards.

How the seismograph works

The seismograph is a **modular digital-telemetry system**:

- **Digital line cable.** A **24-bit digitizer** sits inside a small plastic shell at every channel position along the cable. Each geophone plugs into its own shell, so the signal is turned into numbers right next to the geophone. This avoids the electrical noise that long analog cables pick up.
- **Modules** come with 3, 10, 12 or 24 channels and can be joined, up to 255 channels on one line. Long lines use **power boosters** along the cable.
- **Master unit.** A small box at the end of the line that controls the power supply, the trigger and the data flow. It warns with a sound when it is ready for the shot.
- **Laptop.** Runs the recording software and connects to the master unit through a USB port. Every recording is shown on screen and saved on the laptop as a `.drm` file, which MASW Studio opens directly.

The equipment list

The carrying case holds the master unit, the cables and the small parts. Check that you have everything:

ITEM	WHAT TO BRING	NOTES
Carrying case	Master unit, USB adapter cable (master unit to laptop), power cable with battery clips	Everything is in its own slot in the foam
Digital line cable	Cable reel with the digitizer shells, one per channel	The shell spacing sets the geophone spacing
Power boosters	One per extra cable section on long lines, if supplied	The software lists them when they are connected
Trigger	Trigger sensor and its cable	Placed next to the strike plate; starts each recording
Geophones	One per channel; **4.5 Hz vertical** geophones with spikes	The system works with every kind of geophone; high-sensitivity, low-frequency models see deepest
Laptop	Laptop with the seismograph's recording software, with a free USB port	Charged, with a spare battery or power bank
Power	Charged 12 V battery	Check it the day before
Source	Sledgehammer, **5–10 kg (10–20 lb)**	Heavier hammers reach deeper (see section 2)
Strike plate	Hard plastic (nylon) or aluminium plate, about 20–25 cm (8–10 in) across	Gives a sharp, repeatable hit
Measuring tape	50 m or 100 m (150 ft or 300 ft) tape	Plus pegs to hold it straight
Markers	Survey flags, spray paint or pegs	Mark each geophone position and the hammer point
Elevation (sloping sites)	Level and staff, or GNSS/GPS receiver	Only needed when the ground is not level (section 7)
Field sheet	Printed field sheet (section 10) and a pencil	Write down every file
Safety	Ear defenders, safety glasses, gloves, safety boots, high-visibility vest	The hammer operator needs all of them

Check everything in the office

1. Charge the 12 V battery, the laptop and any GNSS receiver.
2. Install the recording software on the laptop, connect the master unit with the USB adapter cable, and power it from the battery.
3. Start the software and press **Recognize**. The master unit must answer, and its battery voltage must show.
4. Install MASW Studio on the computer you will use for the analysis (section 9). It can be the same laptop or an office computer.

2. Plan the line

Pick the place

- Choose a **straight line** across the area you want to know about. The result describes the ground under the middle of the geophone line.
- Prefer **level, open ground**. A gentle, even slope is fine (section 7). Avoid lines that cross a ditch, a wall, a kerb or a change from soil to pavement.
- Keep away from **noise**: running engines, generators, busy roads, pumps, trees moving in strong wind and people walking near the geophones. If you cannot avoid traffic, record between vehicles.

- Keep at least a few metres from **buried pipes, basements and foundations** if they are not what you are looking for.

How deep can you see?

The depth you can see depends mainly on the **length of the geophone line** (the spread) and on how much low-frequency energy the source produces. A good rule of thumb: you can usually see to about **one-third to one-half of the spread length**.

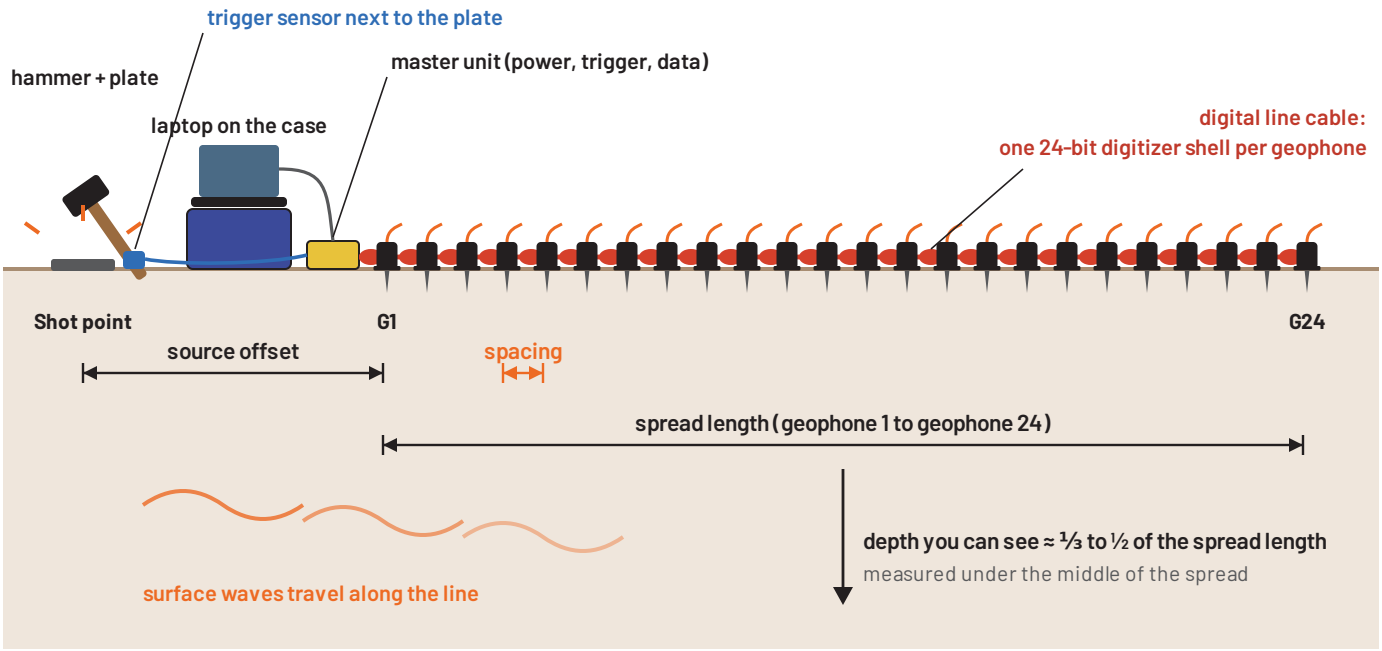
WANTED DEPTH	GEOPHONE SPACING (24 CHANNELS)	SPREAD LENGTH	SOURCE
5–10 m (15–30 ft)	1 m (3 ft)	23 m (75 ft)	5 kg (10 lb) hammer
10–20 m (30–65 ft)	2 m (6.5 ft)	46 m (150 ft)	5–10 kg (10–20 lb) hammer
20–30 m (65–100 ft), e.g. Vs30	3 m (10 ft)	69 m (225 ft)	10 kg (20 lb) hammer, or a weight drop, plus repeated shots

- **Closer spacing** shows more detail near the surface.
- **A longer spread** sees deeper, but the far geophones get weaker signals.
- Vs30 needs the model to reach 30 m. On soft ground that usually needs the longest spread in the table.

Where the hammer goes

The hammer point (the **shot point**) is on the same line, a short distance before the first geophone. This distance is the **source offset**.

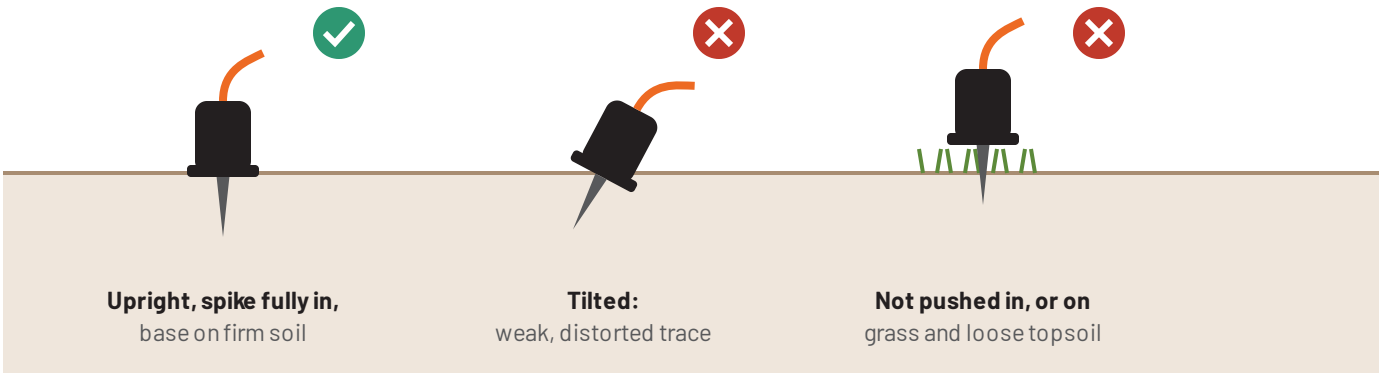
- Use about **one-quarter of the spread length** (5–10 m for a 46 m spread).
- Too close (less than one spacing) and the first geophones are swamped by the hit. Too far and the far geophones hear very little.
- Recording a second shot from the **other end** (the reverse shot) is good practice: it costs five minutes and gives a second, independent profile.



SIDE VIEW OF A MASW LINE: SHOT POINT, MASTER UNIT, DIGITAL LINE CABLE AND GEOPHONES

3. Lay out the line

1. **Stretch the measuring tape.** Lay the tape along the line on the ground and fix both ends. Decide which end is **channel 1** (normally the end nearest the hammer point) and write it on the field sheet.
2. **Mark the positions.** Put a flag or paint mark at every geophone position (for example 0, 2, 4 … 46 m) and at the hammer point.
3. **Unroll the digital line cable** from its reel along the tape, starting with channel 1 at the first mark. Put each digitizer shell at its mark. Keep the cable flat on the ground; a cable moving in the wind adds noise.
4. **Long lines:** connect the next cable section, with a power booster if one is supplied, until the line has all its channels.
5. **Plant the geophones.** At each shell, scrape away grass and loose topsoil, then push the geophone in by hand until its base sits on the ground. Tap the spike in with a small hammer if needed, but never hit the geophone case.
6. **Check they are upright.** A vertical geophone must stand straight (within about 10–20° of vertical, depending on the geophone type; check its data sheet). A tilted or loose geophone gives a weak, noisy trace.
7. **Plug each geophone into its shell.** The shells keep the electronics protected from moisture, but keep the connectors clean and dry.



PLANT EVERY GEOPHONE UPRIGHT AND FIRM; TILTED OR LOOSE GEOPHONES SPOIL THE RECORD

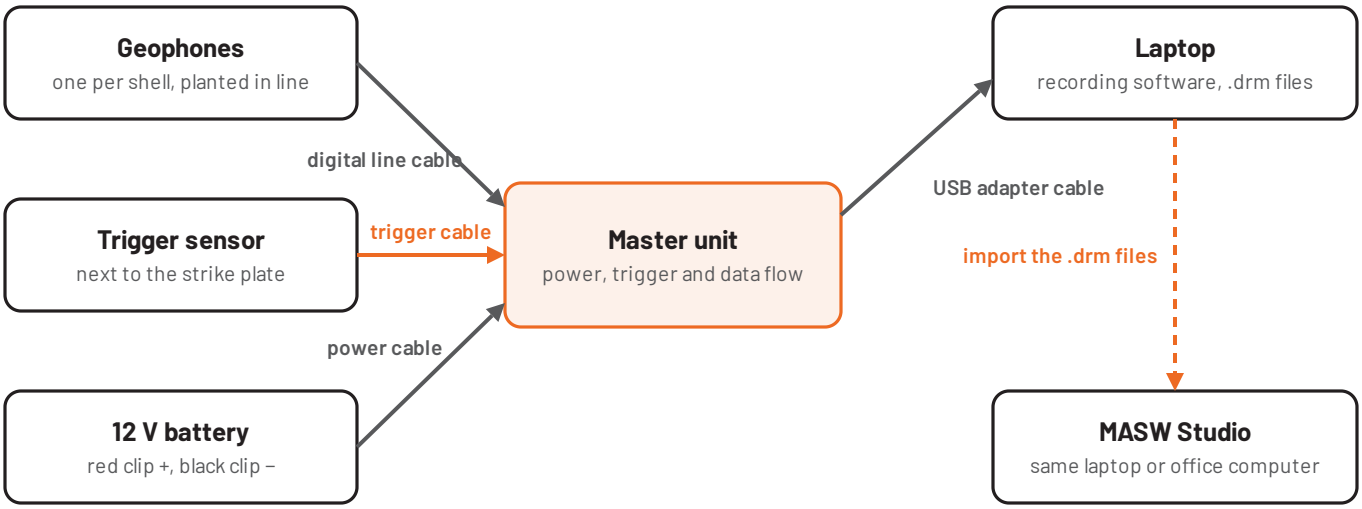
On pavement or concrete you cannot use spikes. Use geophone base plates or tripod stands, or sit the geophones on a small bag of sand. Every geophone must still be firmly in contact with the ground.

4. Connect the master unit and the laptop

Open the carrying case at channel 1's end of the line, then:

1. **Line cable:** plug the digital line cable into the master unit.
2. **Trigger:** connect the trigger sensor cable to the master unit. The trigger sensor goes on the ground next to the strike plate (section 6).
3. **Power:** connect the power cable to the master unit, then clip it to the 12 V battery: red to **+**, black to **–**.
4. **Laptop:** connect the master unit to a USB port on the laptop with the USB adapter cable from the case.

The master unit now controls the power supply, the trigger and the data flow for the whole line.



WHAT CONNECTS TO THE MASTER UNIT

5. Set up the recording software

With a few simple steps you are ready to record. The button names below are those of the recording software shown in the video; they may differ slightly between software versions.

1. **Start the software** on the laptop and connect to the master unit.
2. **Press Recognize.** The software finds every channel connected along the line and, on long lines, the power boosters. **Check that the number of channels is right** (for example 24 / 24) and that the battery voltage is shown. A missing channel means a loose connector between two cable sections or a faulty shell.
3. **Choose the work folder** (**Change Folder**). Every recording is saved there. Use a new folder for each site or line.
4. **Choose the MASW configuration** (the **MASW** button), and save it with **Name and Save** so you can reuse it.

Recording settings

SETTING	USE	WHY
Sampling rate	1,000–2,000 samples per second (1 ms or 0.5 ms)	Plenty for surface waves (they are below about 100 Hz)
Recording time	1 s; 2 s for long spreads or very soft ground	The slow surface waves must reach the last geophone
Pre-trigger / delay	0	Time zero is the moment of the hit; MASW Studio reads recordings without pre-trigger
Channels	All enabled; the same gain on every channel	Disabled channels are skipped in MASW Studio too
Positions	X of every channel and of the **shot location** (Z too on sloping ground)	Saved in the file; MASW Studio reads them on import

Positions and noise check

1. **Enter the positions.** In the channel table, type the X position of each channel along the line (for example 0, 2, 4 … 46 m) and the **shot location** (for example –10 m for a hammer point 10 m before channel 1). Y stays 0 on a straight line. On sloping ground, type each elevation as Z.
2. **Check the noise.** Use the noise monitor, if your software version has one, before the first shot. Every channel should show a small, steady signal. A flat channel is not connected; a channel with large jumps is loose or in the wind.

6. Record a shot

1. **Place the strike plate** on the hammer point and press it firmly onto the ground. Put the **trigger sensor** on the ground right next to the plate.
2. Make sure nobody is walking near the line and no vehicle is passing. The hammer operator wears ear defenders, safety glasses and gloves; everyone else stands back.
3. **Start the recording** in the software and accept or type the file name.
4. **Wait for the beep.** The master unit warns with a sound when it is ready for the shot, and the software shows that it is waiting for the shot.
5. **Hit the centre of the plate** squarely, with a full swing.
6. **Check the record.** The data are downloaded, shown on the screen and saved in the work folder on the laptop. Look at the record (see below), and write the file name, shot location and any notes on the field sheet.
7. **Repeat.** Record three to five shots at the same point; MASW Studio can stack them (Dataset → Stack / split records). Then move the plate to the other end for the reverse shot, if planned, and update the shot location.

> **Want a 2D section?** Either move the whole line along by a few geophone spacings after each shot (a roll-along) and record again, or record one long line with many channels. MASW Studio builds a 2D section from either.

What a good record looks like

- **Good:** every trace shows a clear, slanting band of large, slow wiggles (the surface waves), arriving later at each farther geophone.
- **Dead channel:** one flat trace. Check the geophone plug and the geophone.
- **Noisy channel:** one trace full of fast jitter. Re-plant that geophone and check for wind or someone standing nearby.
- **Clipped:** waveforms with flat tops on the near geophones. Hit a little more softly, or lower the gain of the nearest channels.
- **Weak far traces:** the far geophones show almost nothing. Hit harder, record more shots to stack, or use a heavier hammer.

7. Sloping ground, riverbeds and valleys

Choose level ground when you can. When the geophones cannot all be at the same height (on a slope, across a riverbed or a valley), the waves travel along the ground, which is longer than the flat distance. On a 30° slope the flat distance is 13 % shorter than the real path.

1. Lay the tape **along the ground** and mark positions along the tape as usual.
2. Measure the **elevation of every geophone and of the hammer point** with a level and staff, or a GNSS/GPS receiver. Write them on the field sheet.
3. Type them as **Z** in the recording software's channel table and shot location, so they are saved in every file.

> **Coming in MASW Studio 1.3.3:** elevations are read from the recordings, or entered under **Dataset → Ground elevation**. Distances are then measured along the ground, and the 2D section follows the shape of the ground. Until then, choose the most level line you can.

8. Before you leave the site

- Copy the whole work folder to a USB stick **and** keep it on the laptop: two copies.
- Check the field sheet: every file has a shot location, the spacing, the end of channel 1 and notes.
- Photograph the line from both ends with the geophones in place. Photos can go in the MASW Studio report.
- Note the GNSS/GPS coordinates of both ends of the line.
- Unplug the geophones, roll the line cable back onto its reel, and pack the master unit and cables into their slots in the case. Count the geophones before you leave.

9. Install MASW Studio and analyse the data

Install

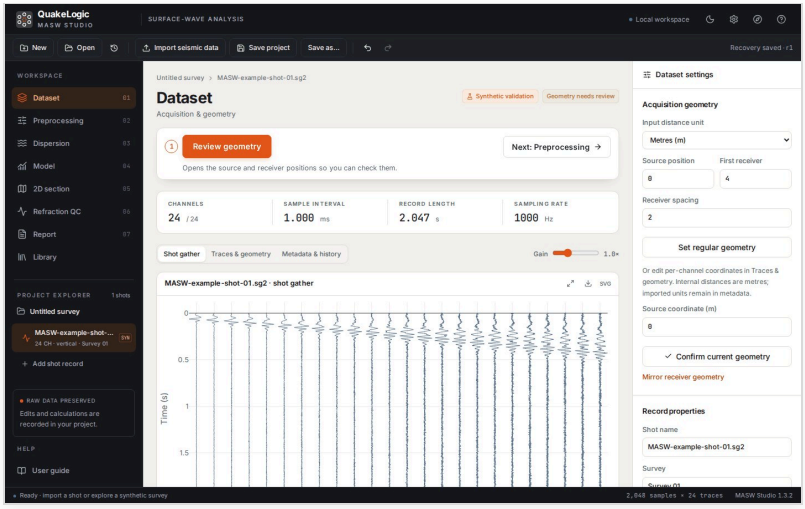
1. Download **MASW Studio Setup** from the QuakeLogic product page, on a computer running Windows 10 or 11 (64-bit).
2. Run the setup file. Windows may show **Unknown publisher**: choose **More info → Run anyway**.
3. Follow the installer and leave **Launch MASW Studio** selected on the last page. The full version opens; there is no licence key and no internet connection is needed.
4. Next time, start it from **Start → QuakeLogic → MASW Studio**.

On first launch the **help wizard** offers a tour. Take it once; it shows every step below on a built-in example recording.

Analyse your recording

The orange button in the bar at the top of each page is always the next thing to press.

1. **Import.** Choose **Import seismic data** and select the `.drm` files from the work folder (several at once is fine). The import summary shows the channels, the sampling and the positions read from the file. Channels disabled in the field stay disabled.
2. **Dataset: check the geometry.** Compare the positions with your field sheet. If they are wrong or missing, enter the source position, first geophone position and spacing under **Set regular geometry**. Then choose **Confirm geometry**.
3. **Preprocessing (optional).** A 5–80 Hz band-pass filter is preset. Preview it, and apply it if the record looks cleaner.
4. **Dispersion.** Choose **Calculate dispersion**. Click along the **bright band** from low to high frequency; six to ten points are enough. The band normally falls from left to right.
5. **Model.** Run the inversion, check that the model curve follows your points, then accept it. Enter the interpretation depth (30 m for Vs30).
6. **2D section.** With several shots along a line, or one long record, build the section.
7. **Report.** Enter client, site and operator, add the site photos, preview the PDF and save it. Choose **Save project** to keep the analysis.



DATASET PAGE: THE ORANGE BUTTON IN THE STEP BAR IS THE NEXT ACTION

The full user guide (in the software under **User guide**, and as a PDF on the product page) explains every setting.

10. Field checklists and field sheet

Before you leave the office

- Carrying case: master unit, USB adapter cable, power cable with battery clips
- Digital line cable on its reel, extra sections and power boosters
- Trigger sensor and cable

- Geophones (one per channel, plus two spares)
- Laptop with the recording software, charged; charged 12 V battery
- Sledgehammer, strike plate, measuring tape, flags, pegs, small hammer
- Level and staff or GNSS receiver (sloping sites)
- Printed field sheets, pencil, safety equipment
- Recognize tested in the office

On site, before the first shot

- Line straight, positions marked and written down
- Channel 1 end decided and noted
- Every geophone upright, firm and plugged into its shell
- Master unit connected: line cable, trigger, battery (red +, black –), laptop USB
- Recognize shows every channel and booster, and the battery voltage
- Work folder chosen; MASW configuration loaded
- Settings: 0.5–1 ms sampling, 1–2 s recording, no pre-trigger, positions and shot location entered
- Noise checked: no dead or noisy channels

Field sheet

FILE	LINE	SHOT LOCATION (M)	CHANNEL 1 AT (M)	SPACING (M)	SHOTS AT THIS POINT	NOTES (NOISE, WEATHER, ELEVATIONS)

11. Troubleshooting

PROBLEM	LIKELY CAUSE	WHAT TO DO
The software cannot find the master unit	USB adapter cable loose, or no power	Re-plug the USB cable; check the battery clips (red +, black –) and charge
Recognize shows fewer channels than the line has	Loose connector between cable sections, missing booster, or a faulty shell	Re-seat the connectors from channel 1 outwards; add the booster on long lines
No beep, or nothing records when the hammer hits	Trigger cable loose, or trigger sensor too far from the plate	Re-plug the trigger cable; put the sensor right next to the plate
Recording starts by itself	Trigger sensor disturbed by footsteps or wind	Keep people away from the sensor; shield it from wind
One trace is flat	Geophone unplugged or broken	Re-plug it into its shell; swap the geophone
One trace is very noisy	Geophone loose, tilted, or in grass or wind	Re-plant firmly and upright
Far traces are almost empty	Not enough energy	Hit harder, record more shots to stack, use a heavier hammer
The bright band fades out below 10–15 Hz	Not enough low-frequency energy, or 10/14 Hz geophones	Heavier hammer, more shots, 4.5 Hz geophones, longer line
MASW Studio says the geometry is missing	Positions were not typed in the channel table	Enter them under Dataset → Set regular geometry
Picks follow a straight rising line	A dashed guide line was picked, not the band	Choose **Reset picks** and click along the bright band
MASW Studio refuses a `.drm` file with pre-trigger	The recording used a pre-trigger or channel delay	Record with pre-trigger 0, or export the file as SEG-2 from the recording software

Help: email support@quakelogic.net from the **Support** menu in MASW Studio, or call QuakeLogic on +1 (916) 899-0391. Email response time: 3 business days.

QUAKELOGIC

MASW STUDIO

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Email response time: 3 business days

Safety first

The hammer operator wears ear defenders, safety glasses, gloves and safety boots. Keep everyone clear of the swing, and never work near traffic without a lookout.

About QuakeLogic:

QuakeLogic Inc. is a California-based engineering and technology company providing innovative, reliable, high-quality solutions to customers worldwide.

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